



Day 23: Relational Database Services (RDS)

[General Interview Questions]

◆ 1. What is Amazon RDS and why is it used?

Answer:

Amazon RDS is a **managed relational database service** that automates **provisioning, patching, backups, and scaling** for popular DB engines.

◆ 2. List the database engines supported by Amazon RDS.

Answer:

MySQL, MariaDB, PostgreSQL, Oracle, SQL Server, and Amazon Aurora.

◆ 3. What is the difference between RDS and Aurora?

Answer:

Aurora is a **high-performance, fully compatible MySQL/PostgreSQL engine** developed by AWS, offering **up to 5x performance** over standard MySQL.

◆ 4. What are the advantages of using RDS over managing databases manually on EC2?

Answer:

- Automatic backups
- Multi-AZ high availability
- Scaling without downtime
- No OS-level management required

◆ 5. What is the purpose of Multi-AZ deployment in RDS?

Answer:

To provide **failover support** and **high availability** by maintaining a synchronous standby replica in another AZ.

◆ **6. What is a read replica in RDS and when would you use it?**

Answer:

A **read-only copy** of your database used for **offloading read traffic** or for **reporting/analytics workloads**.

◆ **7. Can you connect to RDS using the standard database client?**

Answer:

Yes, using the **endpoint, port, and credentials**, you can connect just like any traditional DB.

◆ **8. What is MariaDB and how does it relate to MySQL?**

Answer:

MariaDB is a **community-driven fork of MySQL**, created after Oracle acquired MySQL, and is fully **compatible** with MySQL syntax and tools.

◆ **9. How does RDS handle backup and recovery?**

Answer:

RDS offers **automated backups**, **snapshot creation**, and **point-in-time recovery** within the backup retention window.

◆ **10. What are the types of storage available in RDS?**

Answer:

- General Purpose SSD (gp2/gp3)
- Provisioned IOPS (io1)
- Magnetic (deprecated)

◆ **11. What does “scaling RDS vertically” mean?**

Answer:

Increasing **instance size (CPU, RAM)** or **storage capacity** of the existing DB instance.

◆ **12. What SQL command is used to retrieve data from a table?**

Answer:

`SELECT * FROM table_name;`

◆ **13. Can you enable encryption for an existing RDS DB instance?**

Answer:

No. You must create a **snapshot and restore it** with encryption enabled to convert an unencrypted DB to encrypted.

◆ **14. What is DB parameter group in RDS?**

Answer:

A collection of engine configuration values that are **applied to one or more RDS DB instances**.

◆ **15. How do IAM policies work with RDS?**

Answer:

IAM manages access to the **RDS APIs** (e.g., create DB), but **not database-level users**. For DB-level access, use native DB authentication or IAM DB authentication (for MySQL & PostgreSQL only).